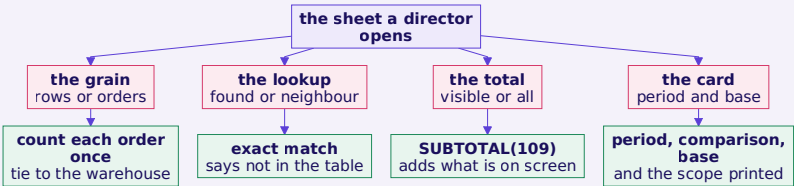


1 The sheet a director opens, and its four silent lies



Each red box prints a plausible number and no error, and the green box under it is the fix. On Friday's exports a Sum over 1,450 payment rows read Rs 39.41 crore against the warehouse's Rs 19.84 crore, C-0195 came back as a neighbour's Rs 16,740, Mumbai's foot read Rs 7,14,890 for Rs 1,56,790 on screen, and a bare Rs 19.84 crore read as a doubled quarter.

**Crux.** Every lie is silent, so every deliverable ships with its check, and one that fails its check is held with its reason.

2 The grain check, and the fix formula

| Check                       | What Friday's raw export showed                 |
|-----------------------------|-------------------------------------------------|
| Rows against ids            | 1,450 rows hold 1,000 orders.                   |
| Total against the warehouse | Rs 39,40,95,490 stands against Rs 19,84,00,000. |
| Remove Duplicates           | It leaves 1,400 rows at Rs 39,40,57,740.        |

```
f flag:=IF(COUNTIF($A$2:A2,A2)=1,1,0)
tree:=SUMIFS(order_amount, segment, "Retail-Plus",
quarter, "Q2", flag, 1)
```

Counted once per order, the tree ties to the rupee: Rs 10.00 crore in Q1, Rs 9.84 crore in Q2.

**Crux.** A pivot is only as honest as the rows under it: say the grain, count rows against ids, tie the total to the warehouse.

3 A lookup has two exits

```
=XLOOKUP(id, ids, revenue, "not in the table")
=IFERROR(INDEX(revenue, MATCH(id, ids, 0)),
"not in the table")
=VLOOKUP("C-0195", table, 5) 4th argument left out
```

XLOOKUP matches exactly by default and shows if\_not\_found for a missing id. VLOOKUP with range\_lookup left out matches approximately, so C-0195 returned C-0194's Rs 16,740 at rank 15. Test every lookup with an id you know is missing.

**Crux.** A lookup that cannot find an id says so; an approximate match answers with a neighbour.

4 The foot of a filtered list

| At the foot      | Rows a filter hid | Rows hidden by hand |
|------------------|-------------------|---------------------|
| SUM              | Added             | Added               |
| SUBTOTAL(9, r)   | Left out          | Added               |
| SUBTOTAL(109, r) | Left out          | Left out            |

Mumbai's 11 members spent Rs 1,56,790 while a SUM foot read Rs 7,14,890. =SUBTOTAL(102, r) counts the visible numbers the foot should be adding.

**Crux.** The foot of a filtered list adds only what the director can see: SUBTOTAL(109), never SUM.

5 The card and its parts

Q2, July to September 2026: Rs 9.84 crore, down 1.6 percent on Q1, April to June 2026 (Rs 10.00 crore).

A change is Q2 minus Q1 over Q1: Retail-Plus is down 29.4 percent, and 41.7 if divided by Q2. A bare Rs 19.84 crore reads as a doubled quarter.

| Scope, a yellow input | The card says                               |
|-----------------------|---------------------------------------------|
| All segments          | Rs 9.84 crore, down 1.6%; 100.0% of revenue |
| All except Business   | Rs 8.15 lakh, down 17.3%; 0.8% of revenue   |
| Retail-Plus           | Rs 4.13 lakh, down 29.4%; 0.4% of revenue   |

**Crux.** One number reaches the front page with its period, its comparison and its base.

6 The operating rule

| Tool          | What it owns                                                  |
|---------------|---------------------------------------------------------------|
| The warehouse | It owns the number, and every join, dedupe and cleaning step. |
| pandas        | It owns the analyst's iteration.                              |
| Excel         | It owns the last mile, and takes what-ifs as labelled inputs. |

The drift check ties the sheet's Q2 total to the warehouse's at every refresh.

**Crux.** The warehouse owns the number, pandas owns the iteration, Excel owns the last mile, and nobody types over the source.

7 The director's what-if

Typed over Q2, five lakh reads down 14.6 percent against Finance's 29.4 until Monday's refresh wipes it, so the what-if goes into a yellow input beside the actual:

| Line     | Retail-Plus, Q2 on Q1 | Where it comes from           |
|----------|-----------------------|-------------------------------|
| Actual   | -29.4%                | The export                    |
| Scenario | -14.6%                | A yellow input of Rs 5,00,000 |

**Crux.** Say yes to the question and no to the edit: the what-if sits beside the actual, never over it.

THE WORDS ON THIS SHEET

- Grain** One row stands for one thing, such as a customer, an order or a payment.

**PivotTable** It summarises a table by the fields you drag in, and changes when refreshed.
- Control total** It is the owner's total, which every sheet built from the data must match.

**First-row flag** A helper puts 1 on each order's first row, so a sum counts it once.
- Exact match** The lookup returns the row whose id equals the one asked for, or nothing.

**Approximate match** The lookup returns the nearest id not above the one asked for, silently.
- if\_not\_found** XLOOKUP's fourth argument says what to show when the id is missing.

**SUBTOTAL(109)** It adds only the visible rows, skipping rows hidden by filter or by hand.